(Please write your Enrollment Number) Enrollment No.________________

Mid Term Examination- ONLINE MODE
(CBCS)
<B.Tech. Ist yr>
(Jan, 2021)

Subject Code:BAS 101	**Subject: Applied Mathematics-I**
Time : 1 ½ Hours	**Maximum Marks :** 30
Note:Q. 1 is compulsory. Attempt any one question from the rest.	

Q1		**(5*3)**
	(a) Find the rank of matrix A defined below by reducing it to normal form $$A=\begin{pmatrix} 1 & 2 & -1 & 3 \\ 3 & 4 & 0 & -1 \\ -1 & 0 & -2 & 7 \end{pmatrix}$$	
	(b) (i) If λ is an eigenvalue of the non-singular matrix , then show that $\frac{1}{\lambda}$ is an eigenvalue of A^{-1}. (ii) $\sum 3^{-n-(-1)^n}$, by root test	
	(c) Test the convergence of the series $$1 + \frac{3}{2}x + \frac{5}{9}x^2 + \frac{7}{28}x^3 + \frac{9}{65}x^4 + \ldots\ldots\ldots\ldots\ldots$$	
Q2		**(7.5,7.5)**
	(a) For what values of λ does the following system of equations possess a nontrivial solution. Obtain the solution for real values of λ $(1-\lambda)\, x + 2y + 3z = 0$ $3x + (1-\lambda)\, y + 2z = 0$ $2x + 3y + (1-\lambda)\, z = 0$	
	(b) (i) Test the convergence of the series $\sum_{n=1}^{\infty} n^2 e^{-n^3}$ (ii) Show that the sequence $\left\{\frac{n}{n^2+1}\right\}$ is monotonic decreasing and bounded. Is it convergent.	
Q3		**(7.5,7.5)**

	(a) Find A^{11} ,where , $A=\begin{pmatrix} -1 & 7 & -1 \\ 0 & 1 & 0 \\ 0 & 15 & -2 \end{pmatrix}$
	(b) (i) Test the series for absolute or conditional convergence $\frac{2}{3} - \frac{3}{4}.\frac{1}{2} + \frac{4}{5}.\frac{1}{3} - \frac{5}{6}.\frac{1}{4} + \ldots\ldots\ldots\ldots.$ (i) Test for convergence of the series, given that $x>0$ $\sum_{n=1}^{\infty} \frac{x^{n-1}}{n.3^n}$

Declaration of the Paper Setter

I have followed these instructions during paper setting with best of my knowledge

a. No direct questions such as definitions, comparisons, diagrams etc has been given where the student can use the book/ online resources directly to answer the question and

b. Ensured that each and every question is verified through google and the same is not directly available and

c. Ensured that the paper covers entire syllabus, all the questions are un- ambiguous, as per the format and followed university norms for setting up the question paper.

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Declaration of the departmental Moderation Committee

	The above paper is moderated and followed above guidelines Name of the HoD: Prof. Chhaya Ravi kant Name of the Faculty1:Dr. Shalini Arora Name of the Faculty2: Dr. Geeta Sachdev